Difference-in-Differences

The Canonical Design: Comparison Groups, Dynamics, Inference

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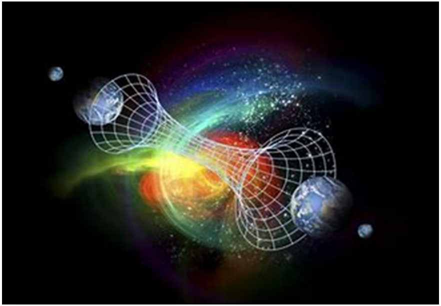
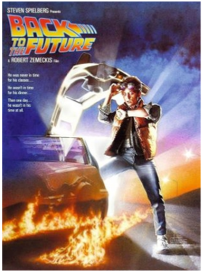


Causal inference and the counterfactual

Difference-in-differences (hereafter **DiD**) is a method of **causal inference** — it identifies a cause-and-effect relationship. It turns on one question: *what would have happened had the shock or policy never occurred?* From that **counterfactual** we recover the effect of the shock or the policy.

Science fiction is built on the same question — stories of an alternative history or a parallel world: *The Man in the High Castle* (P. K. Dick), *Back to the Future*, *Doctor Strange* (Marvel), and many more.



What is difference-in-differences?

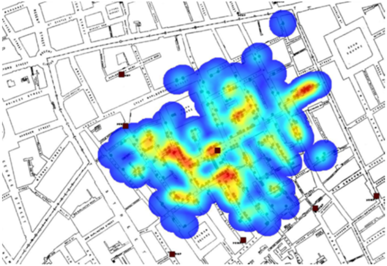
A policy hits **some units** at **some date**. We want its causal effect — but treated and untreated units differ systematically, so neither a treated-vs-control comparison nor a before-vs-after comparison is clean.

The idea in one line

Compare the **change** in the treated group to the **change** in a comparison group. The second difference nets out fixed level differences *and* common time shocks.

- Needs **panel** (or repeated cross-sections): the same groups, before and after.
- **Simple DiD is the TWFE model from the panel chapter** — the only new ingredient is one explanatory variable: a **dummy** for "treated group × after treatment".
- This chapter: the design with **one treatment date**. The next two chapters extend it to **staggered** adoption and to treatments that are a **dose** or that **switch on and off**.

The origin: John Snow and cholera (1855)



Origin: epidemiology — the fight against cholera in London. Cholera was assumed to be transmitted by **air**; John Snow (1855) argued instead that it came from **infected water**.

How to prove it? A *natural experiment*. The whole city drew water from the same point in the Thames, until a reform split the supply: some districts now received water from **downstream** of the sewers (infected), others from **upstream** (assumed uncontaminated).

⇒ *treated vs control*, *before vs after* — the first difference-in-differences.

Roadmap

1. The **canonical** 2×2 — potential outcomes, parallel trends, the estimator.
2. **The comparison group** — what makes a control credible.
3. **Dynamics** — event studies and two-way fixed effects.
4. **Diagnostics** — pre-trends, bounds, placebos, composition.
5. **Inference** — serial correlation, and one treated group.

*Recent synthesis: Roth, Sant'Anna, Bilinski & Poe (2023). Textbook treatment: Cunningham, *Causal Inference: The Mixtape*.*

Section 1

The 2×2

Potential outcomes and the missing counterfactual

Each unit i has two potential outcomes: $Y_i(1)$ **if treated**, $Y_i(0)$ **if not**. The treatment indicator $D_i \in \{0, 1\}$ selects the one we observe, and the causal effect is the gap between the two:

$$Y_i = D_i \, Y_i(1) + (1 - D_i) \, Y_i(0), \qquad \tau_i = Y_i(1) - Y_i(0).$$

The fundamental problem of causal inference

For any unit we observe **only one** potential outcome. A treated unit reveals $Y_i(1)$; its $Y_i(0)$, what would have happened **without** treatment, is the **missing counterfactual**.

- Individual effects τ_i are not identified; the targets are **averages over groups**.
- Notation: $\mathbb{E}[\, \cdot \mid D_i = 1\,]$ is the average **among treated** units, $\mathbb{E}[\, \cdot \mid D_i = 0\,]$ the average **among untreated** units.

The target: the average effect on the treated

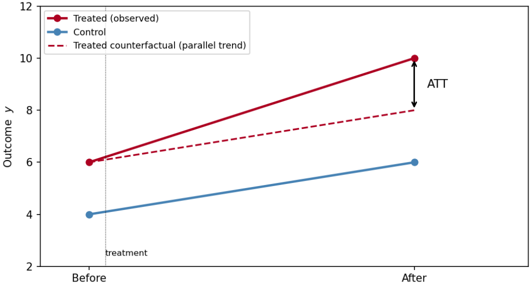
Average treatment effect on the treated (ATT)

$$\text{ATT} = \mathbb{E}[\, Y_i(1) - Y_i(0) \mid D_i = 1 \,].$$

- The effect for the units that **were** treated: did the policy change the outcome of its recipients?
- The average effect on the **untreated**, $\mathbb{E}[\, Y_i(1) - Y_i(0) \mid D_i = 0 \,]$, is a different parameter when effects are heterogeneous; DiD does not identify it.
- **DiD targets the ATT.** The missing piece is the treated group's untreated counterfactual $\mathbb{E}[\, Y_i(0) \mid D_i = 1 \,]$, which **parallel trends** will supply.

The two-by-two: two differences

Two groups, two periods — we now index outcomes by time, $Y_{it}(d)$. DiD builds the treated group's missing counterfactual by assuming it would have moved **like the control**:



$$\widehat{\text{ATT}} = \underbrace{(\bar{y}_{T,\text{post}} - \bar{y}_{T,\text{pre}})}_{\text{treated change}} - \underbrace{(\bar{y}_{C,\text{post}} - \bar{y}_{C,\text{pre}})}_{\text{control change}}$$

Parallel trends: the identifying assumption

Parallel trends

Had the treated group *not* been treated, its outcome would have changed over time by the **same amount** as the control group. Writing $\Delta Y_i(0)$ for that change in the **untreated** outcome:

$$\mathbb{E}[\Delta Y_i(0) \mid D_i = 1] = \mathbb{E}[\Delta Y_i(0) \mid D_i = 0].$$

- **Levels** may differ (fixed effects absorb them); only the **trend** must match.
- **Untestable** (an unobserved counterfactual): defended with pre-period evidence and economic argument, never proved.
- **No anticipation**, the companion assumption: treated units do not respond before the date, so their pre-period outcome is $Y_i(0)$.

The regression form

The 2×2 DiD is one OLS regression:

$$y_{it} = \alpha + \gamma\, D_i + \lambda\, \mathrm{Post}_t + \beta\, (D_i \times \mathrm{Post}_t) + \varepsilon_{it}.$$

- γ — the fixed treated-vs-control gap; λ — the common time shock.
- β **— the DiD coefficient, equal to the ATT** under parallel trends; the interaction $D_i \times \mathrm{Post}_t$ equals d_{it} (one for treated units in the post period, zero otherwise).
- In a **balanced** panel this equals two-way fixed effects (the dynamics section); with entry and exit, the two estimates differ.

Example: Card and Krueger (1994)

New Jersey raised its minimum wage (1992); neighbouring Pennsylvania did not. Compare fast-food employment across the state line.

FTE employment	Before	After	Change
New Jersey (treated)	20.44	21.03	$+0.59$
Pennsylvania (control)	23.33	21.17	-2.16
Difference-in-differences			$+$**2.76**

- Employment in New Jersey **rose** relative to Pennsylvania — not just “did not fall” — against the textbook competitive prediction.
- The cross-border design makes parallel trends plausible: similar labour markets, same macro shocks — the control was chosen for that reason (next section).

Section 2

Comparison group

Choosing the comparison group

Parallel trends is an assumption on the **shocks**: absent the policy, the treated and the comparison group would have received the same ones. The choice of the comparison group is part of the identification strategy and is made on economic grounds before estimation.

1. **Common shocks.** Same industry, markets and geography, so that the time effect f_t is common to both groups. A group on another market follows its own price cycle.
2. **Not exposed to the policy** (SUTVA: one unit's treatment does not change another unit's outcome). Identify the channel of the policy (a price, a quantity, a payment) and verify that it did not move for the comparison group.
3. **No shock specific to the comparison group** during the estimation window.
4. **Observable implications.** The channel (item 2) and the pre-treatment trends (the dynamics section) can be documented; the counterfactual cannot.

Order of the argument

The comparison group is justified first, on items 1–3. Estimation then tests the observable implications of that justification; it does not replace it.

Refine the comparison: region × year effects

With a panel, the common time shock need not be *national*. Replace f_t by $f_{r(i),t}$ — one effect per **region and year**:

$$y_{it} = f_i + f_{r(i),t} + \beta\, d_{it} + u_{it}.$$

- Now treated and control units are compared **within the same region in the same year**: a drought, a regional aid scheme, a local processor's collapse all cancel out.
- The comparison only uses regions where **both** groups are present — a smaller sample, larger standard errors: the cost of a comparison restricted to neighbours.
- A fixed effect **removes** a shock from the comparison; clustering (panel chapter) only accounts for it in the standard error. Put the fixed effect where the shocks are.

Section 3

Dynamics

From the 2×2 to panel TWFE

With many units and periods, write the DiD as **two-way fixed effects**:

$$y_{it} = f_i + f_t + \beta\, d_{it} + u_{it}, \qquad d_{it} = D_i \cdot \mathrm{Post}_t.$$

- f_i absorbs fixed unit differences, f_t the common time shocks (recall the panel chapter). With **common timing** (all treated units adopt at the same date), $\beta = \mathrm{ATT}$ — an equality that **breaks under staggered timing** (next chapter).
- The static β is an **average over a dynamic path**: over a long post period it mixes the first years and the later ones, and it changes with the window. It also says nothing about the **pre-trends**. The event study shows both.
- The **estimation window** is fixed before estimation. Years excluded for a documented shock are named in advance, never after seeing the coefficients.

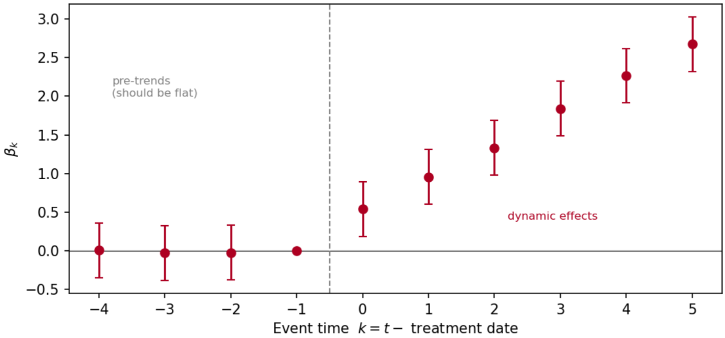
The event-study specification

Replace the single switch by a full set of **leads and lags** around each unit's treatment date:

$$y_{it} = f_i + f_t + \sum_{k \neq -1} \beta_k \mathbf{1}\{t - t_i^* = k\} D_i + u_{it}.$$

- $k = t - t_i^*$ is **event time** (t_i^* = the treatment date); $k = -1$ is the **omitted baseline**, so each β_k is relative to the period just before treatment.
- β_k for $k < 0$ are **pre-trends**; for $k \geq 0$ they are **dynamic treatment effects**.
- With one treatment date this is simply one coefficient per **calendar year** on the treated dummy.

Reading the event study



- **Flat, near-zero** coefficients before 0 are consistent with parallel trends; the post coefficients trace the effect's **dynamics**.
- A sloping pre-period ($\beta_k \neq 0$ for $k < 0$) is a failure of parallel trends before treatment: the comparison group has to be revised.

A caution on pre-trends

Passing a pre-trend test is reassuring, not decisive

Tests of $\beta_k = 0$ for $k < 0$ often have **low power** — they miss real violations — and conditioning on "passing" can itself distort the estimate (Roth, 2022).

- Treat them as **evidence**, alongside an economic case for why the trends should align.
- **Plot** the coefficients; never rely on a single p-value (on event-study design and visualization, see Freyaldenhoven et al. 2021).

Section 4

Diagnostics

Bounds on the violation: Rambachan and Roth (2023)

When the pre-period is *imperfect* rather than broken, ask how large a post-period violation the effect could survive.

The idea

Let M be the largest period-to-period deviation from parallel trends in the pre-period. For a multiplier $\bar{M}$ chosen by the analyst, allow the post-period deviation to be **at most** $\bar{M} \times M$, and compute the confidence interval that remains valid under that restriction.

- $\bar{M} = 1$ allows a violation as large as the worst pre-period year. Report the interval for $\bar{M} = 0.5, 1, 1.5, 2$; the first $\bar{M}$ whose interval contains zero measures the margin.
- Bounds measure how far the estimate is from needing a **better comparison group**; a failed pre-period calls for one, not for a different standard error.
- Implemented as `HonestDiD` (R, Stata). It gives the sentence to report: the result survives violations up to $\bar{M}$ times the pre-period ones.

Placebo tests

A design that identifies an effect where there is one should find **nothing** where there is none:

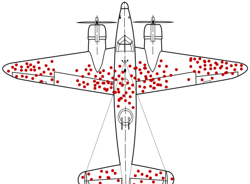
- **Placebo in time** — apply the design to a date at which nothing happened, inside the pre-period.
- **Placebo group** — apply the design to a group that could not have been affected.
- **Placebo outcome** — an outcome the policy cannot reach through the claimed channel.

A placebo in time is a 2×2 on pre-treatment years, a linear combination of the pre-period coefficients: a rejection is a failure of parallel trends at that date. Placebo groups and outcomes test the channel: a rejection means some other shock separates the groups. Both belong in the paper.

Composition: the balanced-panel check

Units enter and leave panels — and **leaving can be an effect** of the policy (firms close, farms exit, workers move).

- In the full sample the post-period mean is computed on **survivors**; if the leavers were the losers, the effect appears to fade faster than it does for any given unit.
- **Check:** re-estimate on units present in every year of a window around the policy (a *balanced* panel).
- Report **both**. They answer different questions: the balanced panel gives the effect on the units that stay, the full panel the effect on the observed population, composition included. The direction of the gap depends on who leaves.



Section 5

Inference

Serial correlation: cluster by unit

d_{it} equals one in **every** post-period of a treated unit: it is perfectly autocorrelated within units, and the residuals u_{it} are persistent too.

Bertrand, Duflo and Mullainathan (2004)

With serially correlated outcomes and a persistent treatment dummy, i.i.d. standard errors reject a *true* null of no effect up to **45 %** of the time at the 5 % level.

- Fix: **cluster by unit** (or by the group at which treatment is assigned) — the sandwich of the panel chapter, block-diagonal within units.
- Report G, the number of clusters; coarser is safer *if* G stays large.

One treated group

Many DiDs have **one** treated group — a sector, a state, a country. Then "cluster at the level of treatment assignment" means $G = 2$: the cluster-robust estimator has one degree of freedom and is uninformative. Clustering by unit ignores the shock common to the whole treated group in a given year (a price cycle, an embargo), which is a **single draw**.

- **What you can do:** cluster by unit and report a coarser level as robustness; run **placebo-in-time** and **permutation** tests (assign the treatment to each control group in turn and see where the true estimate falls in that distribution); Conley–Taber (2011) estimate the distribution of the group-year shock from the control groups, Ferman–Pinto (2019) correct for unequal group sizes.
- **What you must say:** with one treated group, the uncertainty that matters is whether the comparison is right, not the sampling error of a mean. Inference rests on the **design** — the control, the window, the placebos — as much as on the standard error.

Section 6

Applied

Build the treatment variables

Six lines create the treatment variables, the region × year cell, the sample and the clustering.

```
df["lait"] = (df["groupe"] == "lait").astype(int)          # 1
df["post"] = (df["annee"] >= 2015).astype(int)             # 2
df["d"]    = df["lait"] * df["post"]                        # 3
df["reg_an"] = df["region"].astype(str) + "_" \
             + df["annee"].astype(str)                      # 4
base = df[df["groupe"].isin(["lait", "caprins"])]           # 5
cl = {"CRV1": "id"}                                         # 6
```

- **1**, **2** — D_i, 1 for a dairy farm, and Post_t, 1 from 2015 on; `.astype(int)` turns `True/False` into 1/0.
- **3** — $d_{it} = D_i \times \text{Post}_t$: the regressor whose coefficient is the ATT.
- **4** — a text label such as `"22_2015"`, one per region and year; its fixed effect compares dairy and goat farms of the **same region the same year**.
- **5** — keeps the treated group and its control; `.isin` tests membership in a list.
- **6** — standard errors clustered by farm, passed to every regression as `vcov=cl`.

The 2 × 2: by hand, then in regression

Five lines reproduce the table of four means and three regression versions, equal only in a balanced panel.

```
w = base[base["annee"].between(2012, 2016)]                   # 1
tab = w.groupby(["groupe","post"])["res_uta"].mean().unstack() # 2
m1 = pf.feols("res_uta ~ lait + post + lait:post", w, vcov=cl) # 3
m2 = pf.feols("res_uta ~ d | id + annee", w, vcov=cl)          # 4
m3 = pf.feols("res_uta ~ d | id + reg_an", w, vcov=cl)         # 5
```

- **1** — three years before, two years after.
- **2** — the mean outcome by group and period, `unstack()` putting the two periods side by side: the 2 × 2 table. The DiD is the difference of the two row changes.
- **3** — the same number as a regression: `lait:post` is the interaction, its coefficient is β; `lait` and `post` alone are γ and λ.
- **4** — with farm and year fixed effects: each farm compared to itself.
- **5** — with region × year effects instead of year effects: neighbours only.

The event study

Three lines estimate one coefficient per year and read them.

```
sc = base[~base["annee"].isin(range(2007, 2012))]                        # 1
es = pf.feols("res_uta ~ i(annee, lait, ref=2014) | id + reg_an",
              sc, vcov=cl)                                                # 2
es.coef()                                                                 # 3
```

- **1** — drops the years 2007 to 2011, the dairy cycle excluded ex ante; ~ negates the condition.
- **2** — `i(annee, lait, ref=2014)` builds one dummy per year, **multiplied by** the treated indicator, and leaves 2014 out as the reference. The coefficient of each dummy is β_k: the dairy-minus-goat gap in that year, relative to 2014.
- **3** — the coefficients, named `annee::2015:lait` and so on. The years before 2015 are the pre-trend check, the years after trace the effect. The four-digit year can be read out of each name with a regular expression.

Pre-trend test, placebo date, balanced panel

Three groups of lines produce the checks of the chapter.

```
b = es.coef()[pre]; V = es._vcov[np.ix_(idx, idx)]          # 1
W = b @ np.linalg.solve(V, b)                                # 2
p = 1 - stats.chi2.cdf(W, len(b))                            # 3
pl = base[base["annee"].between(2000, 2006)].copy()          # 4
pl["d_pl"] = pl["lait"] * (pl["annee"] >= 2004).astype(int)  # 5
bal = base[base["id"].isin(ids_bal)
           & base["annee"].between(2012, 2018)]              # 6
```

- **1** to **3** — the joint test that all pre-2015 coefficients are zero. `pre` lists their names, `idx` their positions; `b` is the vector of estimates, `V` their covariance block (`np.ix_` selects it); `W = b'V^{-1}b` is the Wald statistic, χ^2 under H_0 with as many degrees of freedom as coefficients.
- **4**, **5** — the placebo: pre-period only, a fake treatment date in 2004; the coefficient of `d_pl` should be near zero.
- **6** — the balanced panel: `ids_bal` lists the farms present every year from 2012 to 2018; the sample keeps them over those years, so the set of farms observed no longer changes.

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